

XRPL Validator — Daily Engineering Report • 2026-08-22

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **8 commits** (xrpl-validator 8)

Engineering changes

- **fix(xrpl-ledger): four fill-precision fixes — six consecutive ledgers byte-for-byte** (b7831e7 , xrpl-validator)
The full-ledger replay now reproduces #106455035-40 exactly (1,019 txs, root == account_hash). Four rules, each ~1 ulp or one nibble, all invisible to key-level gates: - payment: the post-round spend-side fee settlement is REMOVED — every walk fill already debits the sender the GROSS (CLOB move_leg_gross, AMM consume/consume_fib in_gross_rate, DirectHop rates), so the settlement charged the transfer rate a SECOND ...
- **fix(xrpl-ledger): single-path re-arbitration in the bridge controller — seven consecutive ledgers** (436102d , xrpl-validator)
The replay now reproduces #106455035-41 byte-for-byte (1,199 txs). Three rules from #106455041-42's maker-bot cluster: - offer: the bridged controller's direct-pair AMM turn now honors the per-round strand arbitration rippled does every driver iteration (activateNext drops strands whose upper bound misses limitQuality, then setMultiPath(activeStrands.size() > 1) — StrandFlow.h:672-674). A SINGLE-path round stops ...
- **fix(xrpl-ledger): fresh multi-path verdict before the AMM turn — nine consecutive ledgers** (f37bf9d , xrpl-validator)
The replay now reproduces #106455035-43 byte-for-byte (1,648 txs).
- **fix(xrpl-ledger): mulRatio requirements + blind-measurement guards — seventeen consecutive ledgers** (fb798b3 , xrpl-validator)
The replay now reproduces #106455035-51 byte-for-byte (~2,750 txs), and the mulRatio campaign lands whole: histswep 426/426 INCLUDING the l106267220 specimen the first attempt regressed.
- **fix(xrpl-ledger): tec rolls back the AccountTxnID stamp; delivery trim rounds nearest — forty consecutive ledgers** (36feb4f , xrpl-validator)
#106455035-74 now replay byte-for-byte (~5,600 txs).
- **fix(xrpl-ledger): settle the taker once per quality level, not per fill — forty-four consecutive ledgers** (84e17e0 , xrpl-validator)
rippled's flow settles the TAKER once per iteration: the fwd pass's DirectStep debits result.in — accumulated with per-offer STAmount adds — and only the MAKERS are settled per offer (consumeOffer's two sends, BookStep.cpp:900-915). An iteration is ONE quality level (or one AMM slice standing in at a level head): BookStep stops at the level edge so the pass's marginal quality is well-defined, and flow() re-enters. ...
- **fix(xrpl-ledger): a book redeems into the run behind it — forty-six consecutive ledgers** (090b551 , xrpl-validator)
A mixed strand's run saw only its own hops when deciding whether hop 0's issuer charges its transfer rate: qualitiesSrcIssues charges srcQOut exactly when the PREVIOUS STEP redeems, and hop 0 assumed strand-head (prev defaults to Issues). But a BOOK before the run reports DebtDirection::Redeems whenever

ownerPaysTransferFee is off (BookStep.cpp:146), and OwnerPaysFee is a dormant amendment — on mainnet a book ALWAYS ...

- **fix(xrpl-ledger): fwd hops reuse rev-stamped debt directions — fifty consecutive ledgers** (84a026e, xrpl-validator) rippled's DirectStepI::debtDirection returns the CACHE in the forward direction (DirectStep.cpp:492-499) — the direction stamped during the rev pass, when no funds had moved. Our run_fwd re-probed src_redeems from the live sandbox, so a hop that DRAINED its line flipped the next hop's view of it from Redeems to Issues mid-pass, and the redeem-charge vanished.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Aggregated Oracle Feed for Vault Liquidations** (lending) Add a new OracleAggregateSet transaction and ltORACLE_AGGREGATE ledger object (keylet::oracleAggregate(owner, id)) holding a bounded STVector256 (max
- **Atomic Multilateral Netting Settlement** (settlement) Add a ttNETTING_SETTLE transaction carrying sfNettingCycle, a bounded STArray (max 20) of {Account, sfCheckID, sfAmount} entries describing a closed o
- **Attested Generic Messages over XChainBridge** (interop) Extend XChainAddClaimAttestation with optional sfMessageType and sfPayload (bounded Blob, max 1KB) SFields carried alongside the existing value-claim

Appendix — commits

Time	Repo	Commit	Subject
00:26	xrpl-validator	b7831e7	fix(xrpl-ledger): four fill-precision fixes — six consecutive ledgers byte-for-byte
10:57	xrpl-validator	436102d	fix(xrpl-ledger): single-path re-arbitration in the bridge controller — seven consecutive ledgers
12:11	xrpl-validator	f37bf9d	fix(xrpl-ledger): fresh multi-path verdict before the AMM turn — nine consecutive ledgers
16:07	xrpl-validator	fb798b3	fix(xrpl-ledger): mulRatio requirements + blind-measurement guards — seventeen consecutive ledgers
17:32	xrpl-validator	36feb4f	fix(xrpl-ledger): tec rolls back the AccountTxnID stamp; delivery trim rounds nearest — forty consecutive ledgers
19:46	xrpl-validator	84e17e0	fix(xrpl-ledger): settle the taker once per quality level, not per fill — forty-four consecutive ledgers
21:16	xrpl-validator	090b551	fix(xrpl-ledger): a book redeems into the run behind it — forty-six consecutive ledgers
22:13	xrpl-validator	84a026e	fix(xrpl-ledger): fwd hops reuse rev-stamped debt directions — fifty consecutive ledgers